CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 8: **Life Processes in Animals** | Quick revision, one page

Nutrition in humans (steps)

- Ingestion (taking in food) → digestion (breaking food) → absorption (into blood) → assimilation → egestion (removal of undigested food).
- Path of food: mouth → food pipe (oesophagus) → stomach → small intestine → large intestine → anus.

Digestive organs and juices

- Mouth: saliva (salivary amylase) digests starch. Stomach: acid and pepsin act on proteins.
- Liver: largest gland; makes bile that helps in digestion of fats (bile has no enzyme).
- Small intestine: completes digestion; villi absorb food. Large intestine: absorbs water; undigested food leaves as faeces.

Breathing and circulation

- Breathing: inhalation (diaphragm moves down, ribs up) and exhalation (diaphragm moves up, ribs down).
- Gas exchange in alveoli of lungs. Respiration: food + oxygen → energy + carbon dioxide + water.
- Blood: plasma, red blood cells (haemoglobin carries oxygen), white blood cells, platelets.
- Arteries carry blood away from the heart; veins carry blood to the heart; the heart is the pumping organ.

Excretion and other organisms

- Excretion: removal of wastes. Kidneys filter blood to make urine (urea, extra water, salts).
- Amoeba takes food by pseudopodia (false feet); ruminants such as cows chew cud and digest cellulose.

MASTERMIND